

BMJ Open Exploring potential pathogenesis mechanisms of onchocerciasis-associated epilepsy through cerebrospinal fluid analysis of children with complex febrile seizures: a pilot study in Aketi, Democratic Republic of the Congo

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ABSTRACT

Introduction Onchocerciasis, commonly known as river blindness, is a parasitic disease caused by *Onchocerca volvulus* affecting millions predominantly in sub-Saharan Africa. Robust epidemiological evidence points to a clinical relationship between onchocerciasis and epilepsy, a condition termed onchocerciasis-associated epilepsy (OAE). Despite extensive research and various successful elimination programmes over the past decades, the pathogenesis of OAE is still unknown. Current hypotheses propose that *O. volvulus* microfilaria, their excretory-secretory products or the newly discovered filarial *O. volvulus* RNA virus 1 (OVRV1) virus may traverse the blood-brain barrier, triggering seizures or immune responses that result in neurological damage. However, direct evidence of microfilaria or their DNA in cerebrospinal fluid (CSF) or brain tissue remains elusive, likely due to immune-mediated parasite clearance. Additionally, investigations into the potential neurotoxicity of these novel filarial viruses have yet to be pioneered.

Methods and analysis This prospective cohort study will involve 100 ivermectin-naïve children aged 2–5 years, recruited from rural communities in the Aketi health zone, located in the Democratic Republic of Congo. This region is known to be an onchocerciasis-endemic area with a high prevalence and transmission of OAE, despite years of community-directed treatment with ivermectin. Lumbar punctures (LP) will be performed in children presenting with complex febrile seizures according to WHO's paediatric guidelines. CSF samples will be examined for white blood cells, protein levels, glycorrachia, microfilaria, OVRV1 and *O. volvulus* biomarkers. Children will be followed annually, monitoring the development of epilepsy and *O. volvulus* infection. This approach aims to elucidate the presence of *O. volvulus* and OVRV1 in the brain and their role in the pathogenesis of epileptic seizures and the myriad of clinical symptoms observed in OAE.

Ethics and dissemination The protocol has been approved by the Ethics Committee of the University of Kisangani (UNIKIS/CE/KGB/001/2025) and the University of Antwerp (project ID 7323-Edge n/a-BUN

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ The study explores whether complex febrile seizure in an onchocerciasis-endemic setting may represent an early clinical manifestation of onchocerciasis-associated epilepsy (OAE).
- ⇒ Cerebrospinal fluid samples will be analysed for presence of *O. volvulus* biomarkers and the filarial virus *O. volvulus* RNA virus 1.
- ⇒ The number of children with early-onset OAE included in the cohort may be limited.
- ⇒ The follow-up duration may be insufficient to fully capture the long-term risk of epilepsy development after febrile seizures.

B3002025000078). Written informed consent will be obtained from all parents and/or legal guardians of children for whom an LP is considered. Findings will be disseminated at national and international levels via meetings and peer-reviewed open-source publications. Study data will be stored in an open repository.

Trial registration number Pan African Clinical Trials Registry (PACTR202507670131109).

INTRODUCTION

Onchocerciasis, also known as river blindness, is a parasitic disease affecting millions of people throughout endemic regions mainly in sub-Saharan Africa.^{1,2} The Global Burden of Disease Study (2021) found that there are at least 19 million cases of onchocerciasis globally.³ The estimated global age-standardised disability-adjusted life years of onchocerciasis was 15.8 per 100 000 population, excluding onchocerciasis-associated epilepsy (OAE).^{3,4} The aetiological agent, the filarial nematode *Onchocerca volvulus*, is transmitted to humans through the bites of infected blackflies (*Simulium* spp).⁵ *O. volvulus* infection

leads to several clinical manifestations, most notably skin lesions (eg, pruritic dermatitis and nodules) and visual impairment (river blindness).⁶ However, recent studies provide solid epidemiological evidence of the association between onchocerciasis and epilepsy, a condition termed OAE.^{7–10} For instance, OAE appears to cluster in specific families and villages in regions endemic for onchocerciasis near blackfly-infested rivers.¹¹ Furthermore, population-based studies demonstrate a clear correlation between successful onchocerciasis elimination programmes and a reduction in the prevalence of OAE.¹² However, the issue of causality remains unresolved.

OAE typically manifests in previously healthy children aged 3–18 years, with peak onset between 8 years and 11 years.^{13 14} The clinical presentation of OAE encompasses a broad spectrum, including a variety of seizures such as generalised tonic-clonic, absences and nodding seizures, the latter associated with nodding syndrome.^{15 16} It may also present with psychiatric symptoms or as Nakalanga syndrome, which is characterised by growth retardation, physical deformities, endocrine dysfunction and mental impairment.^{17 18} The pathophysiological mechanisms at the root of these heterogeneous symptoms, however, require further research. It has been hypothesised that the microfilaria of *O. volvulus*, the endosymbiotic *Wolbachia* bacteria, or their excretory and secretory products (ESPs) may cross the blood-brain barrier (BBB), either directly triggering seizures or eliciting immune responses that lead to neurological damage and seizures.^{19 20}

Prior to the introduction of community-directed treatment with ivermectin (CDTi)—the basis of the preventive/therapeutic approach for onchocerciasis—*O. volvulus* microfilaria had been detected in the cerebrospinal fluid (CSF) of individuals with onchocerciasis who did not exhibit OAE.^{21 22} Duke *et al* found that diethylcarbamazine (DEC) treatment was associated with an increase of *O. volvulus* microfilaria in the CSF. This phenomenon was hypothesised to be related to the DEC's ability to enhance the BBB permeability.²² However, direct evidence of microfilaria or their DNA in the CSF or brain tissue has been elusive in recent studies. This is possibly due to the clearance of parasites by the host's immune system, as the studies were conducted several years after the onset of symptoms.^{23–25}

Recently, a novel paradigm has been proposed following the discovery of potential neurotropic viruses in nematodes.²⁶ While the impact on the parasite's biology remains unknown, the ability of these viruses to cross specific barriers, such as the BBB of the parasite's host, could contribute to the associated pathology. Through viral metagenomics and transcriptome analysis, two potential pathogenic rhabdoviruses (*O. volvulus* RNA virus 1 (OVRV1) and OVRV2) have been identified in the adult worm and microfilaria of *O. volvulus*.²⁷ However, as OVRV2 was detected exclusively in the L2 larval stage of *O. volvulus*—a maturation stage in the nematode without transmission to humans—its relevance to human disease appears negligible. In contrast, OVRV1 was detected in

the majority of tested *O. volvulus* adult worms, microfilaria and larvae.²⁷

In order to definitively establish the presence of *O. volvulus* and/or associated neurotropic viruses such as OVRV1 in the brain and elucidate their role in the pathogenesis of epileptic seizures, it is imperative to conduct research focusing on the early stages of OAE. Although children generally develop OAE starting at 5 years of age, epilepsy onset in regions endemic for onchocerciasis has been reported to occur as early as 3 years old.^{14 28} Observations in onchocerciasis-endemic areas with high *O. volvulus* transmission indicate that children first become *O. volvulus* infected and later develop epilepsy.²⁹ Furthermore, research shows that epilepsy incidence may increase with higher microfilaria density during childhood and that the age of onset of nodding syndrome may be linked to early childhood exposure to *O. volvulus*.^{18 20 30} Therefore, an aetiological study in a high *O. volvulus* transmission area focusing on children below the age of 5 years, who are still ivermectin-naïve (not yet eligible for CDTi), presents a unique opportunity to investigate the pathogenesis of OAE at the onset of their first seizures.

We propose to perform a lumbar puncture (LP) in children with complex febrile seizures in an area with high ongoing *O. volvulus* transmission. The WHO's paediatric emergency triage, assessment and treatment group suggests performing LP in febrile children with signs of meningitis, <18 months old, with complex seizures, prior antimicrobial use before assessment, those not vaccinated for *H. influenza* type b and/or streptococcal meningitis, or if immunisation status is unknown.³¹ Other studies also suggested that in resource-poor countries, where the incidence of serious central nervous system (CNS) infections is high, a policy of non-selective LP in febrile children who convulse should be recommended.^{32 33}

Globally, febrile seizures are the most common type of acute seizures in children, with a prevalence of 2–7% in individuals before age 5 years.³⁴ Although definitions of febrile seizures vary, the WHO and International League Against Epilepsy (ILAE) use this definition: 'a seizure occurring in childhood after 1 month of age associated with a febrile illness not caused by an infection of the CNS, without previous neonatal seizures or a previous unprovoked seizure and not meeting the criteria for other acute symptomatic seizures'.^{31 35} Febrile seizures are often classified as simple or complex febrile seizures, depending on their characteristics. Simple febrile seizures make up 75% of the attacks (table 1).^{34 36}

After LP, CSF will be examined for the presence of white blood cells (WBCs), protein and glycorachia, microfilariae and other *O. volvulus* biomarkers, with the rationale that some of them might present with an acute phase of OAE. It has been reported that children with a genetic predisposition to (febrile) seizures may have an increased risk of developing OAE¹² and that complex febrile seizures are often predictive of CNS infection.³⁶

Table 1 Clinical characteristics of simple and complex febrile seizures and febrile status epilepticus ^{34 36}		
Simple febrile seizure	Complex febrile seizure	Febrile status epilepticus
Generalised seizure without focal features	Seizure with focal features	Seizure lasting >30 min
Duration <15 min	Duration >15 min, or seizure lasting <15 min but requiring anticonvulsant medication for termination	Recurrent or serial seizures without recovery of consciousness between seizures, with a total duration >30 min
No recurrence within 24 hours	Recurrence within 24 hours	
No pre-existing neurological abnormality	Pre-existing neurological abnormality	
	Postictal neurological abnormality (eg, Todd's paralysis)	

Hypothesis

Some of the children with complex febrile seizures may present with an acute phase of OAE, during which indicators of *O. volvulus* presence and/or OVRV1 in the brain may be detectable in the CSF. The newly identified OVRV1 virus present in the *O. volvulus* worm could play a role in the pathogenesis of OAE.

Main study aim

To provide solid evidence of the presence, or lack thereof, of microfilaria, OVRV1, ESPs and other *O. volvulus* biomarkers in the CSF of children with early-stage OAE.

Study objectives

This study aims to elucidate the role of the parasite in triggering epileptic seizures. Understanding these mechanisms could lead to better diagnostic tools and targeted

interventions, ultimately contributing to improved health outcomes for affected people in endemic regions.

METHODS

Study design

Prospective cohort study.

Study setting

The Democratic Republic of Congo (DRC) is home to multiple onchocerciasis-endemic foci with a high prevalence of epilepsy.^{30 37–39} One of those is the Aketi health zone in the Bas-Uélé province (figure 1). This health zone encompasses rural communities where agriculture, hunting and fishing constitute the main economic activities of the residents.

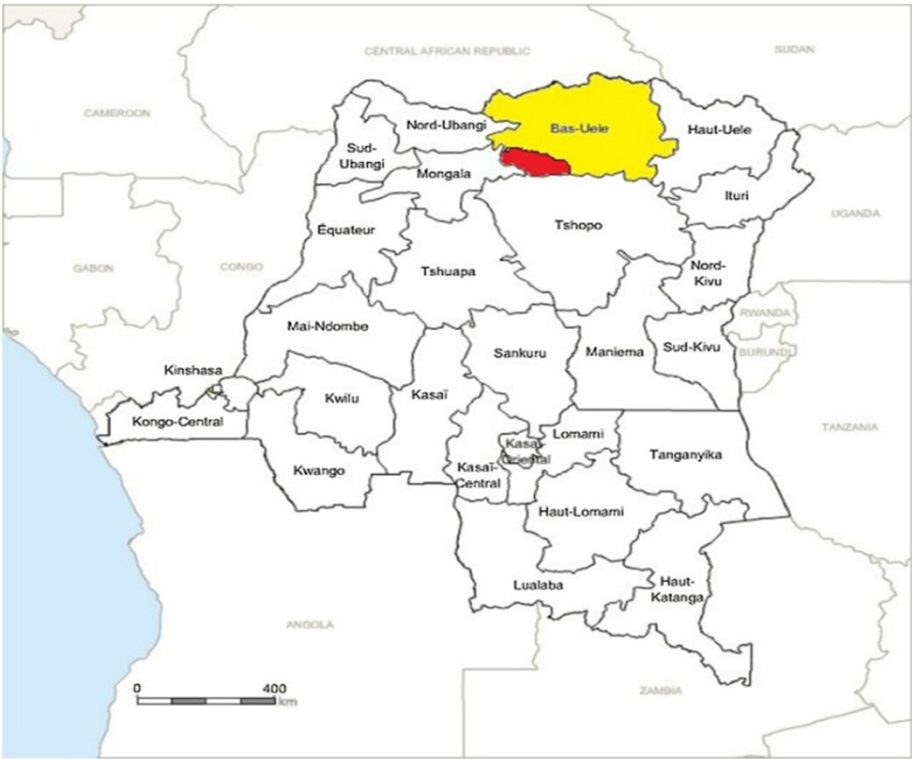


Figure 1 The Aketi health zone (in red) located in the province of Bas Uélé (in yellow) in the Democratic Republic of Congo.³⁷

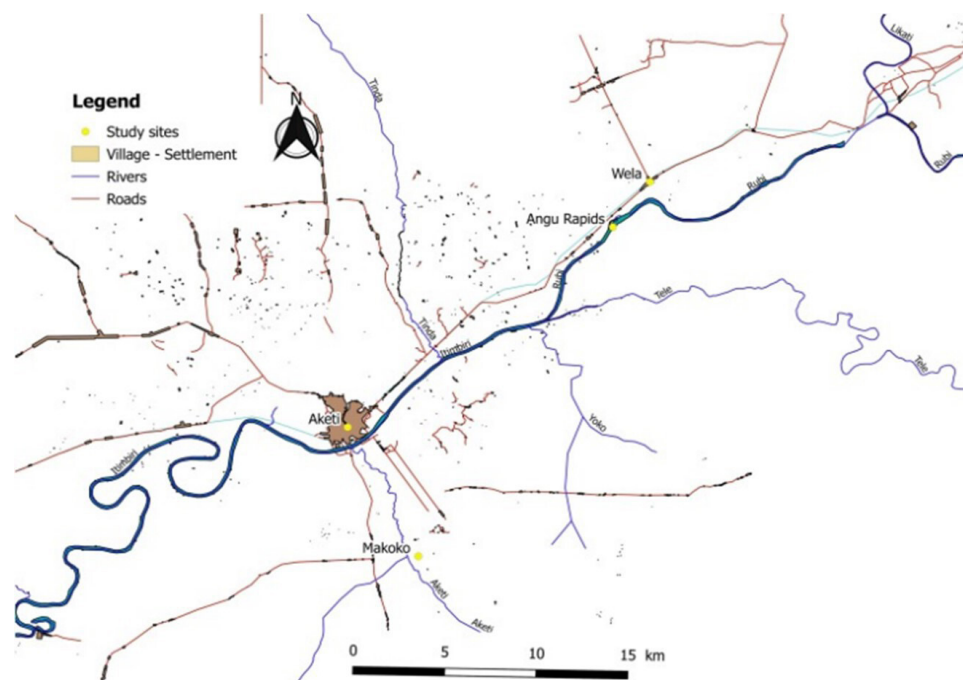


Figure 2 The villages of Aketi, Makoko and Wela and the Itimbiri and Rubi rivers, located in the Aketi health zone.³⁷

The region features several fast-flowing rivers (figure 2) that provide suitable breeding grounds for blackflies (*Simulium* spp), the onchocerciasis vector, subsequently translating into high blackfly biting rates.

The morbidity pattern is primarily characterised by poverty-related conditions, with infectious diseases and epilepsy constituting the primary causes for consultations and hospital admissions.³⁷

Despite the implementation of a CDTi programme for the control of onchocerciasis since 2003, the prevalence and incidence of epilepsy and onchocerciasis skin lesions remain high.^{37 39} In 2015, in a house-to-house study, the prevalence of epilepsy was 6.8% in Wela village and 8.4% in Makoko village.³⁹ In 2017, the epilepsy prevalence in Aketi town was 5.7%, with nine persons with epilepsy (PWE) (13.8%) experiencing head nodding seizures.³⁷ Furthermore, the high prevalence of OV16 antibodies in children is indicative of high ongoing onchocerciasis transmission.³⁷ Moreover, in Aketi, an association was observed in persons with epilepsy between high *O. volvulus* microfilaria density and increased seizure frequency.³⁰

Time schedule

The project will run for 4 years. The major milestones for this study will be spread over time as follows:

- ▶ Study preparation: 6 months.
- ▶ Enrolment period: 2 years.
- ▶ Statistical analysis and community follow-up: 3 years (overlapping with enrolment period).

Sample size calculations

Considering the exploratory nature of this study (no preliminary data available), we will recruit as many children as possible during the 2-year enrolment period of

this pilot phase. A target of 100 children has been set for this pilot study.

Study population and samples to be collected

Children aged 2–5 years old will be recruited into the study due to multiple factors. To investigate the pathogenesis of OAE, participants need to be studied before the onset of symptoms, which typically manifest starting at the age of 5 years.⁹ Additionally, as there is a lack of pharmacokinetic evidence confirming the safety of ivermectin in children under 5 years old, this age group is ineligible for ivermectin treatment.⁴⁰ Participants must be at least 2 years of age as the reproduction cycle of *O. volvulus* takes 2 years minimum (assuming that the earliest childhood infection happens during the neonatal period).¹

The second criterion is the presence of complex febrile seizures, as defined by the WHO and ILAE criteria.^{31 35} Also, children between 2 years and 5 years who present with fever and seizures or other neurological disorders for which the healthcare personnel deem that a LP is warranted will be recruited.

Study procedures

Selection and enrolment of the study participants

Ivermectin-naïve children aged between 2 years and 5 years in onchocerciasis foci with complex febrile seizures or other neurological disorders that warrant CSF investigations will be recruited.

Exclusion criteria

1. Previous ivermectin intake.
2. Persistent altered mental status, coma and/or focal neurological signs.
3. Bleeding disorder.

4. Infection at the site of the LP.

Interview and examination

For each study participant, pretested questionnaires will be administered to the parents and/or legal guardians. This questionnaire will be used to document relevant information about the child (sociodemographics, seizure history, ivermectin use, (pre)term birth features, nutritional status, OAE in mother, etc). On obtaining informed consent from the parents and/or legal guardians, the children will be clinically examined to evaluate if they match the aforementioned eligibility criteria. After confirmation, they will be asked to participate in the study, and 2 mL of venous blood will be collected. Recruitment will be consecutive.

Lumbar puncture

Local healthcare workers in the participating hospitals and clinics will be trained on how to collect CSF (via LP) and process it according to standard protocols, with extra care so that the first few drops of CSF that emerge after a LP are not collected to minimise contamination by microfilaria from the skin. LP will only be performed if indicated and prescribed by the treating physician, and the left-over CSF after hospital workups will be stored by the study team for further processing.

CSF processing and analysis

The initial CSF analysis will focus on parameters of diagnostic value for further treatment such as WBC count with formula, CSF protein, CSF glucose, CSF serum/glucose ratio and gram stain following the hospital's protocols and available facilities. Children will be treated for bacterial meningitis in the presence of the following criteria: CSF white cell count of >5 per μL and predominance of neutrophils, protein of $>0.8\text{g/dL}$, CSF glucose of $<1.0\text{mmol/L}$ with or without bacteria seen on Gram stain.

Consent will be obtained from the parents or legal guardians to use the remaining CSF for future research purposes. The samples will be aliquoted and stored immediately at -20°C or in liquid nitrogen if this is available. While maintaining the cold chain, samples will be shipped to the University of Antwerp (), where they will be stored at -80°C . In Antwerp, CSF will be re-examined for the presence of WBC and formula, proteinorrhachia and glycorrhachia, microfilaria and other *O. volvulus* biomarkers such as OVRV1 reverse transcription polymerase chain reaction (RT-PCR). In case we detect OVRV1 in CSF samples, we will assess the capacity of this virus to infect neural cells and determine its neurotropism and neurotoxicity. Blood samples will be used for OV16 and OVRV1 antibody testing. OVRV1-positive CSF samples collected in liquid nitrogen will be seeded on human induced pluripotent stem cell-derived neurospheroid models.⁴¹

Follow-up

Children will be followed up annually for a maximum of 3 years to verify whether they develop epileptic seizures.

When they develop epileptic seizures, they will be treated with antiseizure medication and when they become eligible for ivermectin treatment (at 5 years of age), ivermectin treatment will be administered.

Study outcomes and data analysis plan

Incidence rate ratios and the 95% CI for epilepsy onset will be calculated in children with evidence of *O. volvulus* or OVRV1 infection in CSF at recruitment, compared with those without CSF infection.

Data handling and storage

All data will be handled and stored according to ethical guidelines and regulations. Data will be stored in a secure, access-controlled database with physical and electronic security measures. Only selected personnel will have access to the sensitive data, further safeguarding participant information.

Quality control

Data will be collected using the REDCap electronic data capture system, enabling real-time data validation, range checks and consistency checks. Principal investigators at the UA will conduct continuous remote monitoring to ensure data completeness and accuracy.

All study personnel will receive standardised training on questionnaire administration, clinical assessment, LP procedures, sample handling and data entry. Standard operating procedures will be used across all study sites.

A mid-study monitoring visit will be conducted to review recruitment procedures, adherence to LP and CSF handling protocols, data quality and consent documentation. Interim descriptive analyses will be performed to identify inconsistencies or protocol deviations.

No predefined stopping rules have been established, as this is an exploratory pilot study aiming to enrol as many eligible children with complex febrile seizures as possible. Early termination of the study will be considered only in the event of serious protocol violations, persistent quality control issues or safety concerns identified by the investigators or ethics committees.

Patient and public involvement

The study was prepared in collaboration with the local non-governmental organisation Adicon representing the community of Aketi. Prior to the study, communities will be sensitised and mobilised about the importance of preventing OAE by participating in the CDTi programme. The study rationale, objectives and procedures will be explained during interactive community meetings. Study findings will be communicated to the village communities as the project advances.

ETHICS AND DISSEMINATION

This study will be conducted in accordance with good clinical practices, the Declaration of Helsinki and applicable regulatory requirements. The protocol has been approved by the Ethics Committee of the University of

Kisangani and the UA. Written informed consent will be obtained from all parents and/or legal guardians of children for whom an LP is considered. During the informed consent procedure, potential participants will be informed about the study in simple language. The risk of performing a LP will be explained and the availability of insurance protection will be mentioned. Of note, LP will only be performed if it is medically relevant for the care of the child and the project will collect any left-over CSF after the tests requested by the treating physician have been conducted. Data will be pseudonymised and encrypted to protect personal and health information. Participating parents and/or legal guardians will be given feedback about the study results. Findings will be disseminated at national and international levels via meetings and peer-reviewed open-source publications. Study data will be stored in an open repository.

DISCUSSION

While strong epidemiological evidence exists surrounding the interplay between onchocerciasis and epilepsy, no concrete pathophysiological mechanisms capable of explaining this association have been established yet. There are unmet needs for aetiological research into OAE, especially concerning the early-stage pathogenesis. This study attempts to tackle one aspect of this enigma, namely, the question of whether microfilaria, OVRV1, ESPs and other *O. volvulus* biomarkers are present in the CSF and responsible for neuroinflammatory/neurotoxic reactions at the core of OAE.

However, several significant challenges must be addressed a priori. First of all, the risks and ethical concerns surrounding LPs. Performing LP in children poses significant risks, including morbidity and potential death from herniation, and can only be performed by trained health personnel. Parents may be reluctant to allow their children to participate due to concerns regarding the perceived pain and invasiveness of a LP, despite its routine application in clinical practice and correct indication given the complex febrile seizures. To address these concerns, the study will cover the costs associated with the LP procedure and the hospital analysis of the CSF.

Additionally, during community follow-up of the participating children, ivermectin will be administered from the age of 5 years following guidelines of the local elimination programmes. Due to the anthelmintic effect of ivermectin, it is unlikely that these children will develop OAE, potentially creating a biased data collection. To counter this, it is essential to enrol mainly young children, aged 2–3 years old, to obtain a sufficient number of ivermectin-naïve children at the end of the study.

In low-resource areas, the lack of adequately trained healthcare personnel, limited infrastructure and insufficient laboratory facilities to process CSF specimens efficiently add serious medical, operational, ethical and financial challenges. Given these challenges, a

multicentric approach is advisable in order to enhance the study's feasibility and statistical power. Other high *O. volvulus* transmission zones with a high prevalence and incidence of epilepsy to consider are areas in the DRC, South Sudan and the Central African Republic. However, in several potential sites in South Sudan, such as Maridi⁴² and Mvolo,⁴³ the incidence of OAE is expected to decrease because of the implementation of bi-annual CDTi. A study site in an onchocerciasis-endemic area in Cameroon could also be considered, but recently the incidence of epilepsy has decreased because of many years of CDTi.^{44–46} Healthcare workers in participating hospitals and clinics will need to receive training on LP and CSF collection and process it according to standard protocols. CSF samples will need to be shipped to the university of Antwerp in Belgium for OVRV1 RT-PCR analysis. Organising such a multicentric study in resource-limited areas requires substantial coordination and resources, which may be difficult to secure and maintain throughout the study period. For all these reasons, a pilot study is planned in Aketi in the DRC, where the local principal investigator, Dr Kungu Gbulu R, is currently already performing LPs in children with complicated febrile seizures to exclude treatable meningitis.

If *O. volvulus*-related biomarkers are not detected in the CSF of children with febrile seizures, the study sample size may be expanded to include an additional 100 children with febrile seizures. These additional participants will be recruited either in the Aketi health zone or at an alternative study site located in an area with high ongoing *O. volvulus* transmission. Any expansion of recruitment to a new site will be subject to additional ethical approval and site-specific feasibility assessments.

In addition, subject to separate ethical approval, the research team may develop a complementary protocol to investigate CSF obtained through postmortem LP in children from onchocerciasis-endemic areas who died following epileptic seizures. Such a study would aim to explore neuropathological and virological markers of *O. volvulus* infection and would be conducted in strict accordance with local regulations, cultural considerations and internationally accepted ethical standards, including informed consent from parents or legal guardians.

FUTURE PERSPECTIVES

Despite these challenges, this study holds the potential to advance our understanding of OAE and contribute to the global effort to eliminate onchocerciasis and associated morbidities, improving health outcomes in affected and at-risk communities. We hope to provide substantial evidence of *O. volvulus*, OVRV1 or derivative products in the CSF. Furthermore, this study will be among the first to investigate the potential neurotoxicity of filarial viruses, potentially leading to the identification of novel biomarkers and the development of new diagnostic tools for elimination campaigns.

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Contributors RC and JNSF designed the study, RC is the guarantor, KGR as main local investigator provided advice about how to practically conduct the study, FSD provided paediatric advice, DWB and CR wrote the first draft, all authors reviewed the paper and contributed to the final version of the paper.

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